

Yu Liu

Phone: +86 18920481010; +852 53930355

E-mail: leni.yuliu@gmail.com; yu.liu@link.cuhk.edu.hk

EDUCATION

The Chinese University of Hong Kong 2022.08 - 2025.10

Faculty of Medicine, Department of Medicine and Therapeutics, Division of Neurology

Ph.D. in Medical Sciences

Beihang University 2018.09 - 2021.06

Postgraduate recommendation

School of Biological Science and Medical Engineering

M.phil. in Biomedical Engineering, GPA: 3.80/4.00

Core courses: Statistics, Biomechanics, Detection and processing of biomedical signal, Modeling and simulation of biological system, Anatomy and physiology, Ergonomics

Thesis: Hemodynamic study of vertebrobasilar dolichoectasia

Hebei University of Technology 2014.09 - 2018.06

Honor Graduate with Distinction, Rank: 1/41

School of Electrical Engineering

B.Eng. in Biomedical Engineering, GPA: 3.71/4.00

Core courses: Linear algebra, Engineering mathematics, Fundamental Theory of Circuit, Signal and linear system, Automatic control theory, Digital signal processing, Digital image processing, Medical imaging system, Fundamentals of mono-chip computers and applications, Biomedical electronics, Biomedical sensor and measurement

Thesis: Non-contact heart rate detection based on photoplethysmographic imaging (**Best thesis award, top 5%**)

WORK EXPERIENCE

Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences 2023.09-2023.11

Visiting Student, Part-time

Cooperation Project: The pathogenic and recurrent mechanisms of symptomatic intracranial atherosclerotic stroke (sICAS): an interdisciplinary study based on advanced neuroimaging and computational fluid dynamics (CFD)

Fosun Aitrox, Shanghai 2021.06-2022.06

Research and Development Engineer, Full-time

Project: Hemodynamic algorithm for fraction flow reserve computed tomography (FFRct)

China National Institute of Standardization, Beijing 2019.08-2019.10

Research Intern, Part-time

Project: Standardized and digital management of medical insurance policy

PUBLICATIONS

Journal (Working in progress)

- **Is Invasive Fractional Flow Measurement Accurate in Intracranial Stenosis? A Computational Simulation Study**
Yu Liu*, Zhengzheng Yan*, Ziqi Li, Yuying Liu, Sze Ho Ma, Bonaventure YM Ip, Thomas W Leung, Jia Liu†, Xinyi Leng†

Under Review

- **Symptomatic atherosclerotic middle cerebral artery stenosis: associations of proximal arterial geometry with plaque location and focal hemodynamics**

Yu Liu, Shuang Li, Xuan Tian, Yuying Liu, Ziqi Li, Tingjun Liang, Haipeng Liu, Linfang Lan, Bonaventure YM Ip, Hui Fang, Bo Song, Yuming Xu, Thomas Leung, Xinyi Leng

Prepare to submission

Journal

- **More severe cerebral small vessel disease associated with poor leptomeningeal collaterals in symptomatic intracranial atherosclerotic stenosis**

Yuying Liu, Shuang Li, Xuan Tian, Jill Abrigo, Bonnie YK Lam, Jize Wei, Lina Zheng, Yu Liu, Ziqi Li, Tingjun Liang, Bonaventure YM Ip, Thomas W Leung, Xinyi Leng

Journal of Cerebral Blood Flow & Metabolism (JCBFM), 2024 (**Q1, IF: 4.9**)

- **Clinical implications of haemodynamics in symptomatic intracranial atherosclerotic stenosis by computational fluid dynamics modelling: a systematic review**

Yu Liu, Shuang Li, Haipeng Liu, Xuan Tian, Yuying Liu, Ziqi Li, Thomas W Leung, Xinyi Leng

Stroke and Vascular Neurology (SVN), 2024 (**Q1, IF: 4.4**)

- **Cerebral hemodynamics and stroke risks in symptomatic intracranial atherosclerotic stenosis with internal versus cortical borderzone infarcts: a computational fluid dynamics study**

Shuang Li, Xuan Tian, Bonaventure Ip, Xueyan Feng, Hing Lung Ip, Jill Abrigo, Linfang Lan, Haipeng Liu, Lina Zheng, Yuying Liu, Yu Liu, Karen KY Ma, Florence SY Fan, Sze Ho Ma, Hui Fang, Yuming Xu, Alexander Y Lau, Howan Leung, Yannie OY Soo, Vincent CT Mok, Ka Sing Wong, Xinyi Leng, Thomas W Leung

Journal of Cerebral Blood Flow & Metabolism (JCBFM), 2024 (**Q1, IF: 4.9**)

- **Hemodynamic significance of intracranial atherosclerotic disease and ipsilateral imaging markers of cerebral small vessel disease**

Lina Zheng, Xuan Tian, Jill Abrigo, Hui Fang, Bonaventure YM Ip, Yuying Liu, Shuang Li, Yu Liu, Linfang Lan, Haipeng Liu, Hing Lung Ip, Florence SY Fan, Sze Ho Ma, Karen Ma, Alexander Y Lau, Yannie OY Soo, Howan Leung, Vincent CT Mok, Lawrence KS Wong, Yuming Xu, Liping Liu, Xinyi Leng, Thomas W Leung

European Stroke Journal (ESJ), 2024 (**Q1, IF: 5.8**)

- **Effects of abnormal vertebral arteries and the circle of Willis on vertebrobasilar dolichoectasia: a multi-scale simulation study**

Yu Liu*, Xinmiao Zhang*, Yawei Wang†, Wentao Feng, Jing Jing, Zhujun Sun, Bitian Wang, Yongjun Wang, Yubo Fan†

Clinical Biomechanics (Clin Biomech), 2023 (**Q3, IF: 1.4**)

- **Effects of stent shape on focal hemodynamics in intracranial atherosclerotic stenosis: A simulation study with computational fluid dynamics modeling**

Haipeng Liu, Yu Liu, Bonaventure Y M Ip, Sze Ho Ma, Jill Abrigo, Yannie O Y Soo, Thomas W Leung, Xinyi Leng

Frontiers in Neurology (Front Neurol), 2022 (**Q3, IF: 2.7**)

- **Variations of human cerebral and ocular blood flow during exposure to multi-axial accelerations: A mathematical modeling study**

Weipeng Li, Bitian Wang, Yawei Wang†, Xiaoyu Liu, Wentao Feng, Tianya Liu, Zhujun Sun, Yu Liu, Songyang Liu, Yubo Fan†

Conference

- **Associations of ICA Bifurcation Geometry, MCA Geometry and Plaque Location in Symptomatic Atherosclerotic MCA Stenosis**
Yu Liu, Shuang Li, Linfang Lan, Xuan Tian, Yuying Liu, Ziqi Li, Haipeng Liu, Bonaventure Ym Ip, Thomas Leung, Xinyi Leng
Poster presentation, European Stroke Organisation Conference (ESOC) 2024
- **Association between Adjacent Arterial Geometry and Plaque Features in Symptomatic Atherosclerotic Middle Cerebral Artery Stenosis**
Yu Liu, Shuang Li, Xuan Tian, Yuying Liu, Ziqi Li, Haipeng Liu, Bonaventure YM Ip, Thomas W Leung, Xinyi Leng
Poster presentation, Asia Pacific Stroke Conference (APSC) 2023
- **Cerebral hemodynamics by computational fluid dynamics modeling in symptomatic intracranial atherosclerotic disease and the clinical relevance: a systematic review**
Yu Liu, Shuang Li, Xuan Tian, Yuying Liu, Haipeng Liu, Thomas Leung, Xinyi Leng
Poster presentation, European Stroke Organisation Conference (ESOC) 2023

* indicates equal contribution.

† indicates corresponding authorship.

PROJECT

- **The pathogenic and recurrent mechanisms of symptomatic intracranial atherosclerotic stroke (sICAS): an interdisciplinary study based on advanced neuroimaging and computational fluid dynamics (CFD)**

National Natural Science Foundation of China (NSFC)/ Research Grants Council (RGC) of Hong Kong Joint Research Scheme, 2021-2023 (Participant)

Affiliated institution: Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences/The Chinese University of Hong Kong

- **Impact of blood pressure variations on focal and global cerebral hemodynamics in symptomatic intracranial atherosclerotic stenosis (sICAS): a pilot study with patient-specific computational fluid dynamics (CFD) modeling**

Health and Medical Research Fund (HMRF), 2018-2022 (Participant)

Affiliated institution: The Chinese University of Hong Kong/ Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences

- **Intelligent Circulatory System Rehabilitation Device Based on Extracorporeal Counterpulsation (ECP) Technology**

National Key Research and Development Plan, Ministry of Science and Technology of the People's Republic of China, 2020-2023 (Participant)

Affiliated institution: Sun Yat-sen University/Beihang University

CHALLENGE

First prize, National Biomedical Engineering Innovation Design Competition for College Students (**top 10%**)

Project: Low-powered and remoted electrocardiogram (ECG) monitoring system

Third prize, iCAN Innovation Contest

Project: Intelligent wheelchair based on real-time pose transformation

HONORS&AWARDS

Research postgraduate scholarship , the Chinese University of Hong Kong	2022-2025
First-class scholarship , Beihang University (3 times)	2018-2021
Outstanding graduate , Hebei Province (top 5%)	2018.06
Outstanding graduate , Hebei University of Technology	2018.06
First-class scholarship , Hebei University of Technology (2 times)	2017-2018
Outstanding student leader , Hebei University of Technology	2017.12
Outstanding student , Hebei University of Technology	2016.12